

Xiling Li

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RESEARCH INTERESTS My research focuses on secure and private data management and its applications including verifiable query evaluation, oblivious outsourced querying and privacy-preserving machine learning, and works extensively with secure multiparty computation, zero-knowledge proofs and differential privacy.

EDUCATION

Ph.D. Computer Science, Northwestern University Jun 2021 - Present

- Advisor: Dr. Jennie Rogers

M.S. Computer Science, University of Washington Dec 2020

- Advisor: Dr. Martine De Cock
- Thesis: *Privacy-Preserving Filter-based Feature Selection with Secure Multiparty Computation*

B.S. Computer Science, University of California, San Diego Dec 2016

RESEARCH EXPERIENCE

Research Assistant, Northwestern University Jun 2021 - Present

- [ZKSQL](#) (VLDB 2023) - 50+ Citations: Proposed the first work [2] on verifiable and efficient query evaluation with zero knowledge proofs for ad-hoc SQL queries in an operator-at-a-time fashion.
- [RESCU-SQL](#) (VLDB 2023 demo): Proposed the first pragmatic OLAP system [1] with all-but-one malicious security for ad-hoc SQL queries.

Research Intern, Dolby @ATG Experience Delivery Jun 2026 - Sep 2026

- Meedle project: Proposed a recall-aware meet-in-the-middle hybrid query optimization.

Research Intern, Dolby @ATG Experience Delivery Sep 2025 - Dec 2025

- Prototyped scalable multimodal lakehouse for heterogeneous AI workloads.

Research Assistant, University of Washington @PPML Group Sep 2019 - May 2021

- [ICML 2021](#) - 80+ Citations: Proposed the first general secure multiparty computation protocol [3] for FFS with malicious security and honest majority, and instantiated feature score protocol with MS-GINI.
- [UbiTtention 2020 Workshop](#) (UbiComp-ISWC 2020): Proposed Mean-Split Gini Impurity algorithm (MS-GINI) [4] for Filter-based Feature Selection (FFS).

TEACHING EXPERIENCE

Guest Lecturer

- Memory Hierarchy*, COMP_SCI 213, Northwestern University Spring 2026
- Relational Algebra*, COMP_SCI 339, Northwestern University Spring 2024
- Database Architecture and Query Evaluation*, COMP_SCI 339, Northwestern University Fall 2023

Teaching Assistant

- COMP_SCI 213: Intro to Computer Systems*, Northwestern University Spring 2026
- COMP_SCI 339: Intro to Database Systems*, Northwestern University Spring 2023-2025

INDUSTRIAL EXPERIENCE

Data Scientist, IBM @Watson IoT Jan 2018 - Aug 2019

- Built a knowledge graph with Neo4J for a reasoning module in a disaster prevention platform.
- Integrated an image pre-processing module with OpenCV into a vision system for manufacturing.
- Developed a millisecond-level face recognition module using FaceNet for warehouse management.

Android Developer, Shenzhen Das Intellitech Co., Ltd @R&D Department Jul 2017 - Dec 2017

- Developed an Android module for door access control by voice within an intelligent building system.

SERVICES **Reviewer:** ICML 21-25; NeurIPS 21-25; ICLR 22-25; AISTATS 25-26; AAAI 26-27; Asiacypt 25

OPEN SOURCE ARTIFACTS **Xiling Li**, Chenkai Weng, Yongxin Xu, Xiao Wang, Jennie Rogers. *ZKSQL: Verifiable and Efficient Query Evaluation with Zero-Knowledge Proofs*. <https://github.com/vaultdb/zksql>, Feb 2023.

TECHNICAL SKILLS	C++, Python, EMP-toolkit , PyTorch, MP-SPDZ , AWS EC2, Ubuntu, Docker
INVITED TALKS	Efficient Oblivious Database Joins , <i>Northwestern Database Reading Group</i> , May 2025. ZKSQL , <i>Northwestern Database Reading Group</i> , Jan 2025. Privacy + Machine Learning , <i>Northwestern AI Journal Club</i> , Nov 2021.
SELECTED PUBLICATIONS	[1] Xiling Li*, Gefei Tan*, Xiao Wang, Jennie Rogers, Soamar Homs. <i>RESCU-SQL: Oblivious Querying for the Zero Trust Cloud</i>. In Proceedings of the VLDB Endowment (PVLDB), Volume 16, No. 12, 4086-4089, 2023. DOI: https://doi.org/ 10.14778/3611540.3611627. [2] Xiling Li, Chenkai Weng, Yongxin Xu, Xiao Wang, Jennie Rogers. <i>ZKSQL: Verifiable and Efficient Query Evaluation with Zero-Knowledge Proofs</i>. In Proceedings of the VLDB Endowment (PVLDB), Volume 16, No. 8, 1804-1816, 2023. DOI: https://doi.org/10.14778/3594512.3594513. [3] Xiling Li, Rafael Dowsley, Martine De Cock. <i>Privacy-Preserving Feature Selection with Secure Multiparty Computation</i>, In Proceedings of the 38th International Conference on Machine Learning, PMLR 139:6326-6336, 2021. [4] Xiling Li, Martine De Cock. <i>Cognitive load detection from wrist-band sensors</i>. In Adjunct Proceedings of the 2020 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2020 ACM International Symposium on Wearable Computers (UbiComp-ISWC '20). ACM, New York, NY, USA, 456–461. DOI: https://doi.org/10.1145/3410530.3414428.